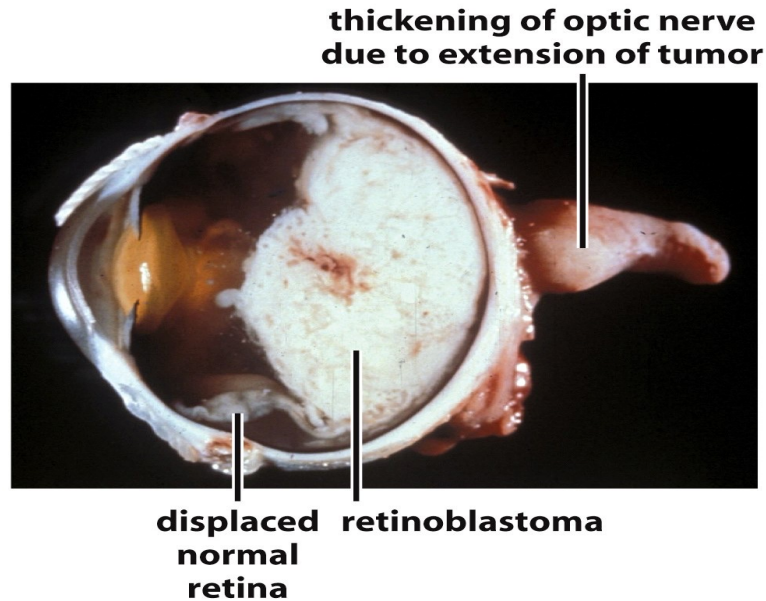


Cancer genetics II



LEUKOCORIA, a white reflex in the pupil

Genetics of tumor suppressor genes



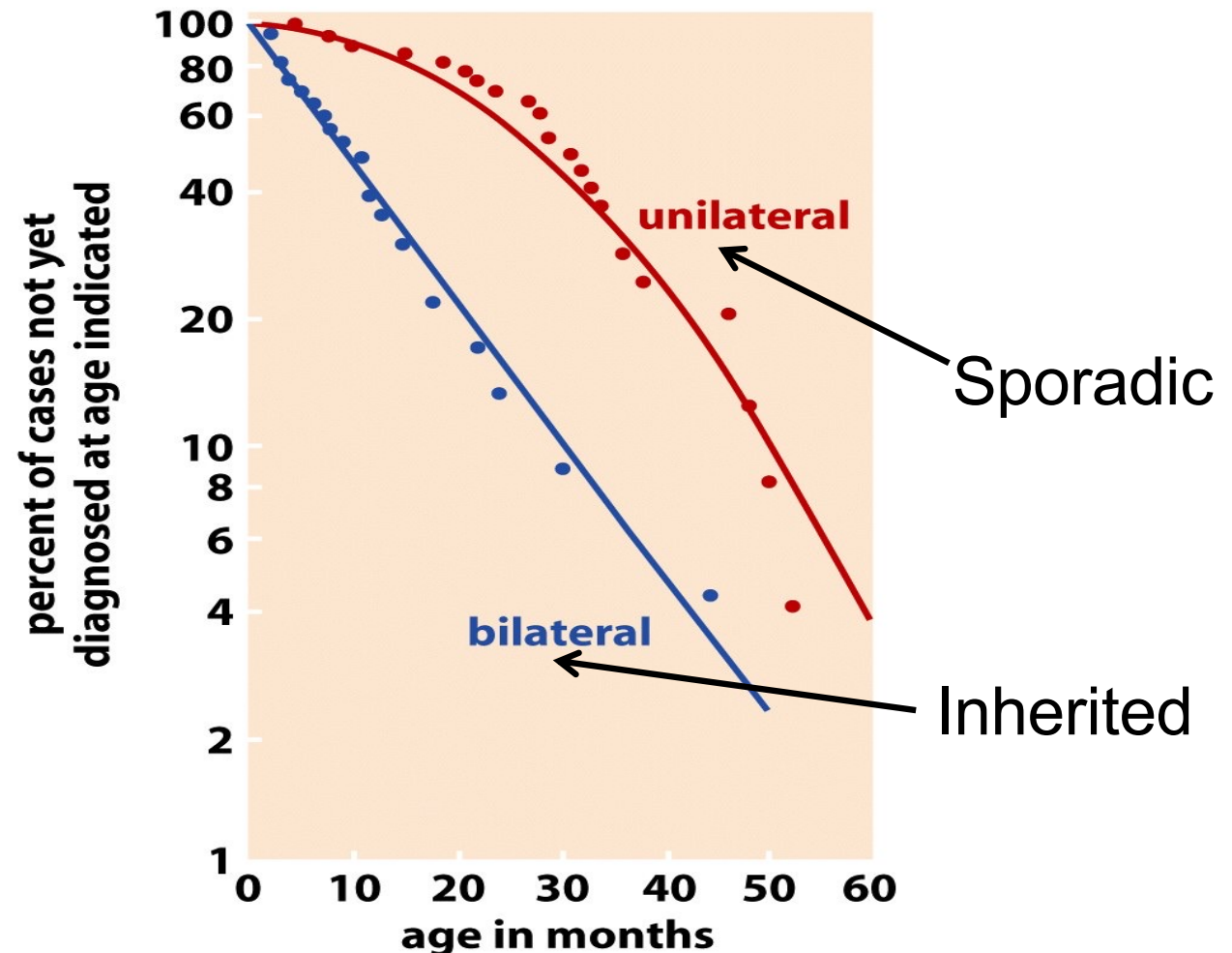
Retinoblastoma is a rare (1 / 15,000) childhood cancer of the eye.

Pedigree analyses show that 40 % of the cases involve an inherited (familial) mutation.

Retinoblastoma may be unilateral (affecting only one eye) or bilateral (affecting both eyes).

The two-hit hypothesis:

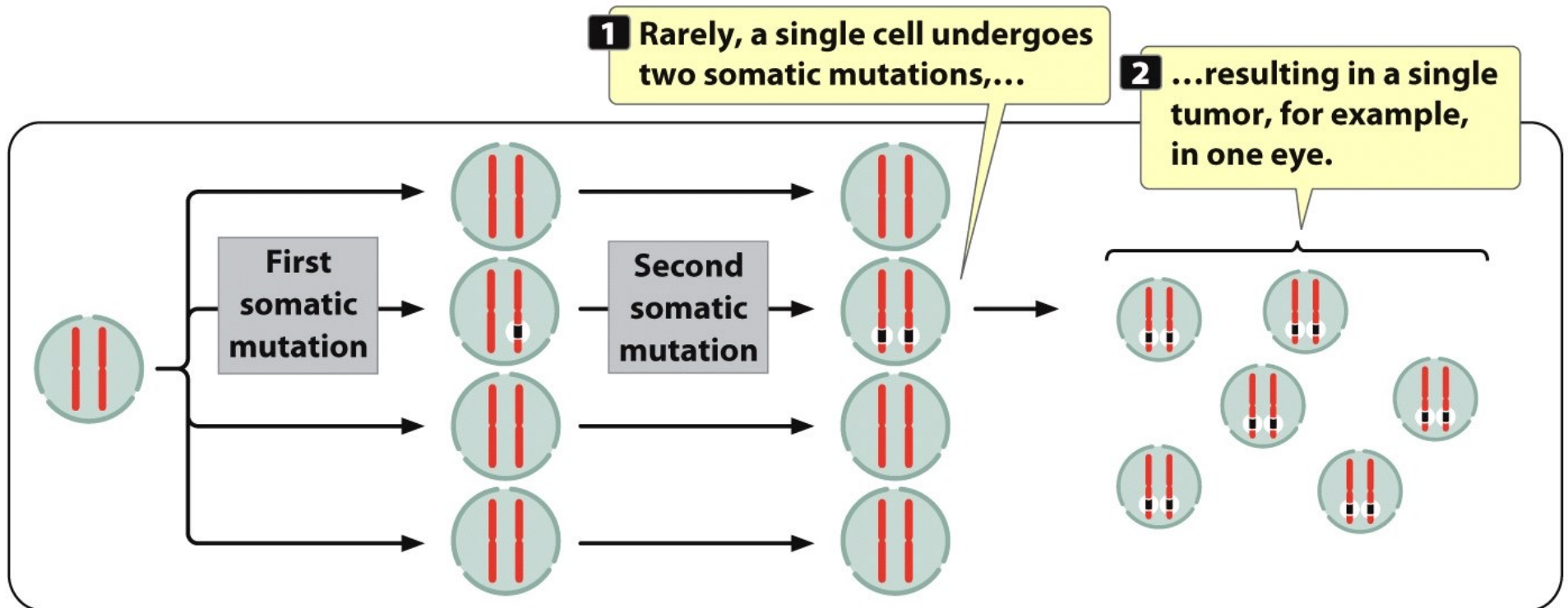
In 1971, Alfred Knudson observed a difference in the kinetics of retinoblastoma development between unilateral and bilateral cases



The two-hit hypothesis:

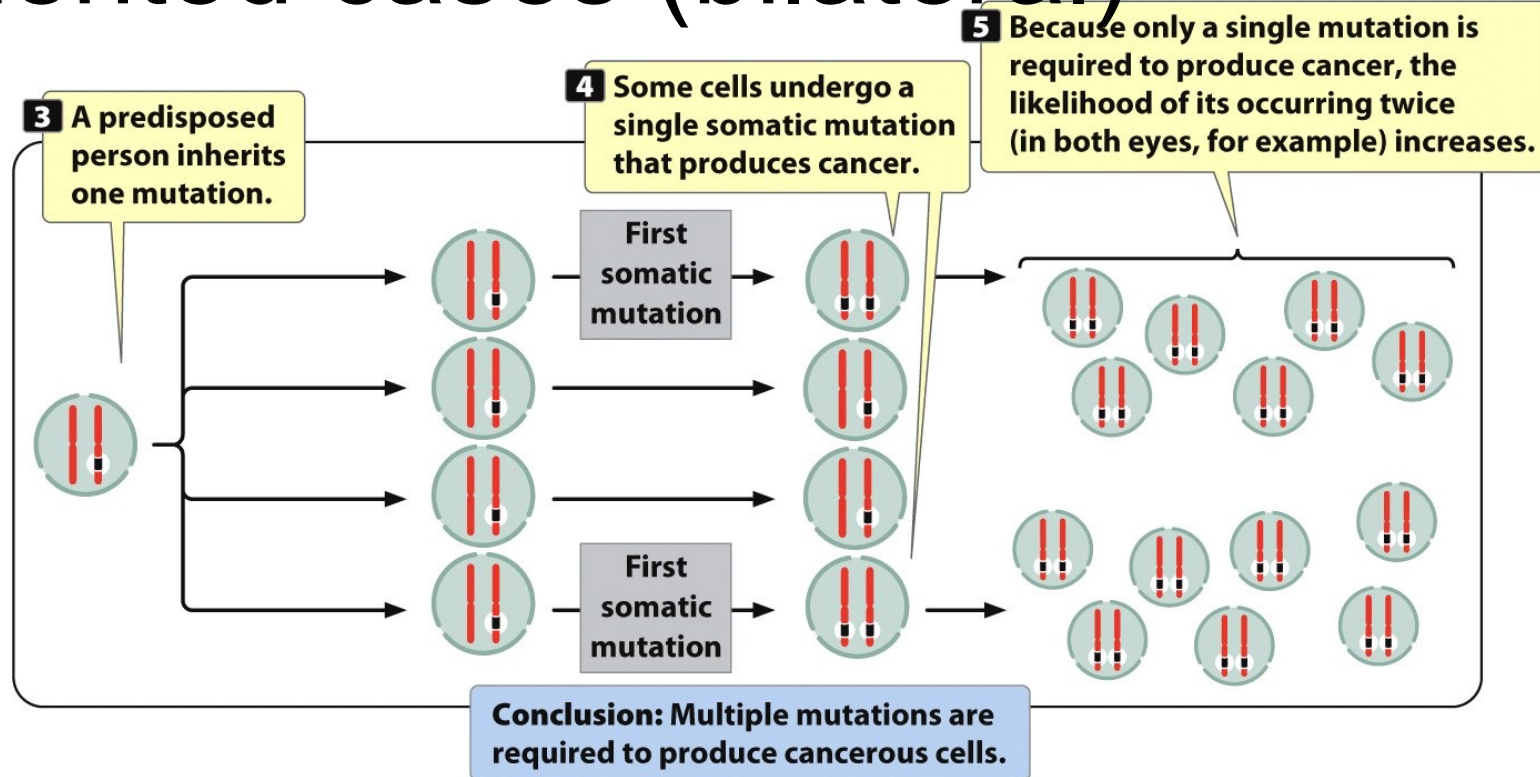
Knudson proposed that retinoblastoma results from two separate genetic defects (mutations), two hits.

Sporadic cases (unilateral)



In sporadic cases, “two hits” must happen to a single cell. Such event is rarer and takes longer to develop.

Inherited cases (bilateral)



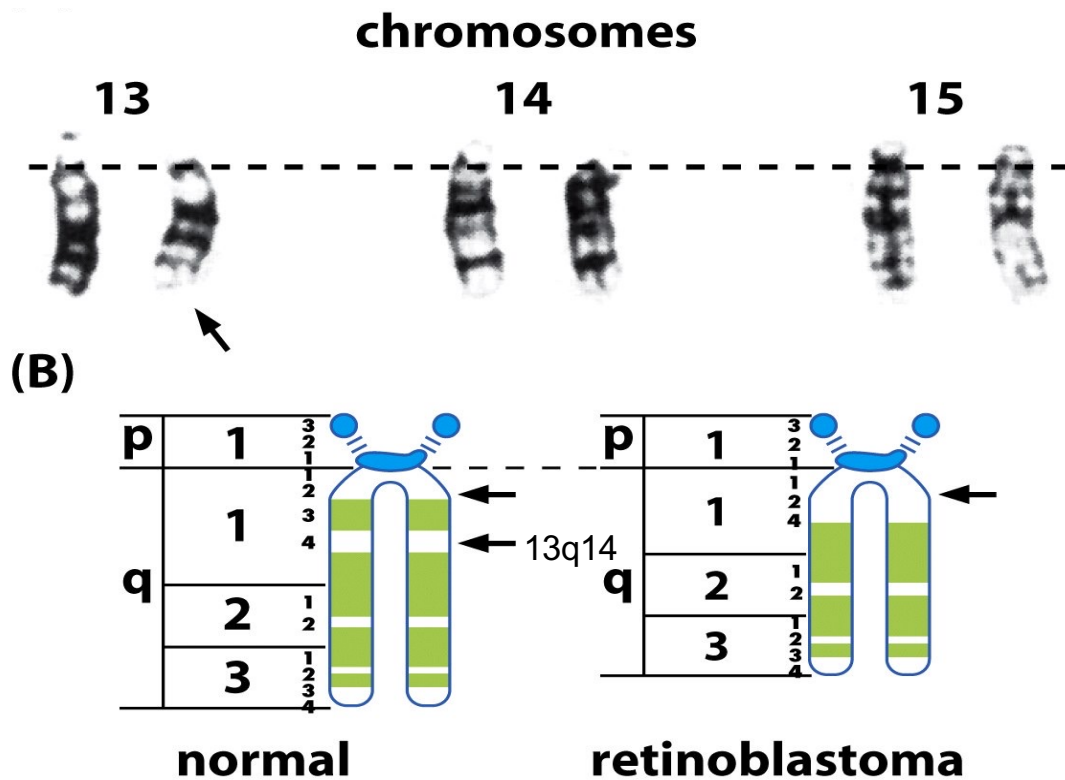
In inherited cases, children are born with the first hit. Chances of having the second hit is high, and it could happen to more than one cell.

The hypothesis is consistent with the idea of having recessive mutations in both alleles of a single gene.

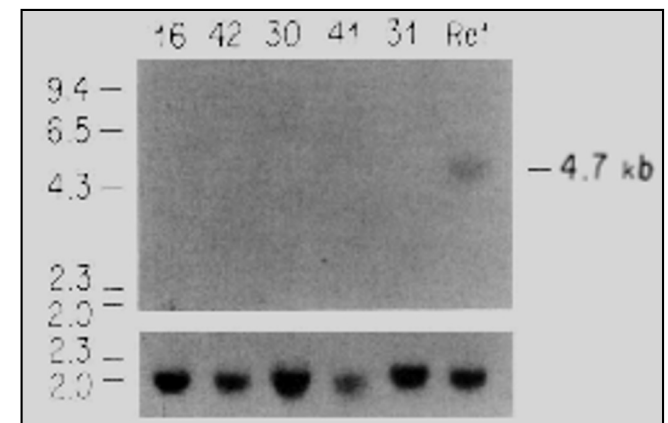
Pierce Figure 23.3

Confirmation of “two hit hypothesis” and existence of tumor suppressor genes

Chromosomal abnormality frequently found in Retinoblastoma



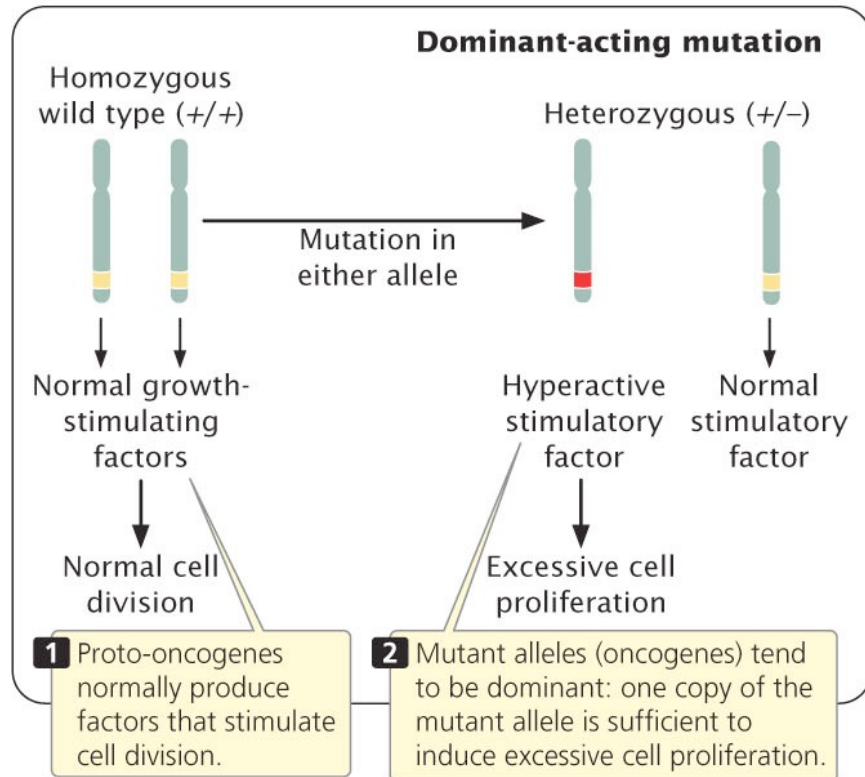
1986: The RB1 gene is cloned, discovery of the first tumor suppressor gene.



(Nature 1986)

Oncogenes and tumor suppressor genes

(a) Oncogenes



(b) Tumor-suppressor genes

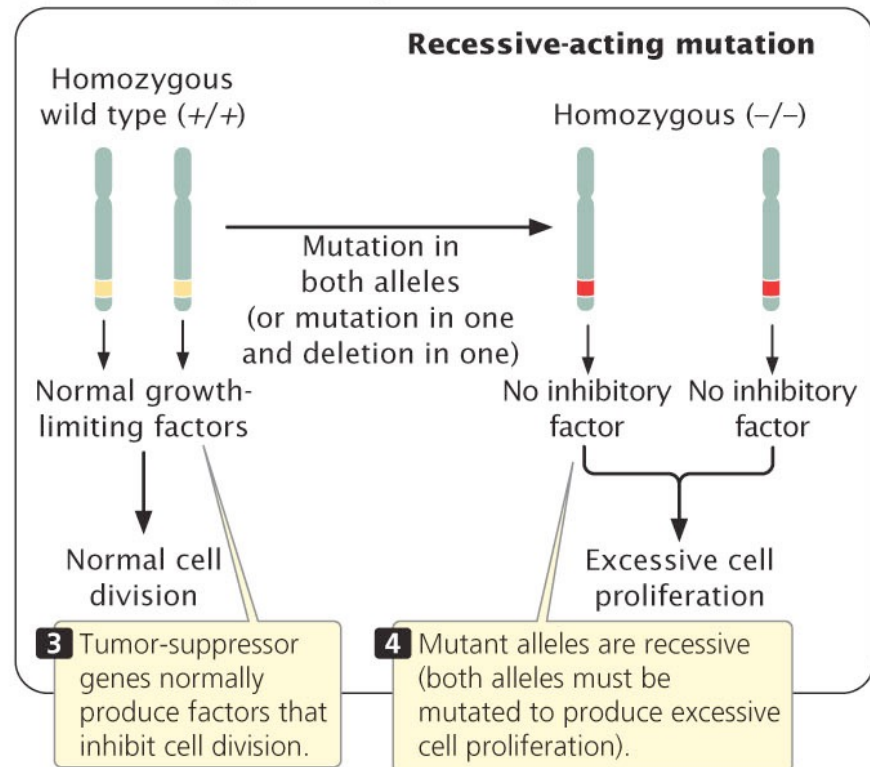
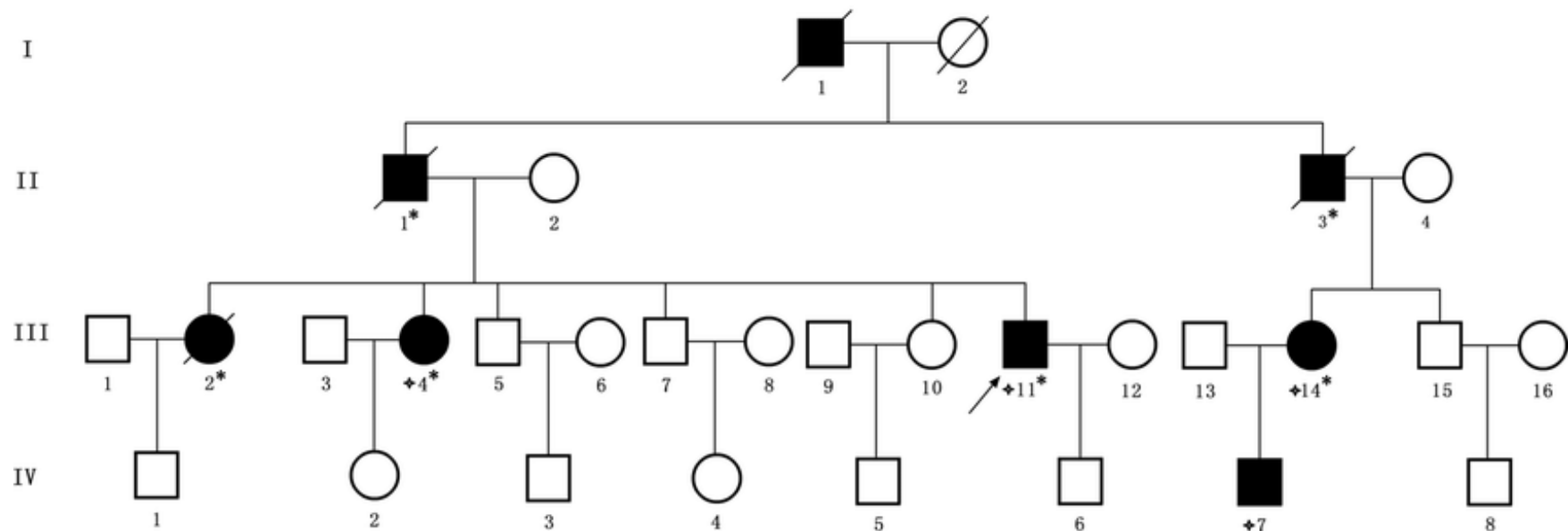


TABLE 23.5 Some tumor-suppressor genes and their normal functions

Gene	Normal Function	Cancer in Which Gene Is Mutated
<i>APC</i>	Scaffold protein, interacts with microtubules	Colorectal
<i>BRCA1</i>	DNA repair, transcription factor	Breast and ovarian
<i>CDKN2A</i>	Regulates cell division	Melanoma
<i>NF1</i>	GTPase activator	Neurofibromatosis
<i>p53</i>	Regulates cell division, apoptosis, DNA repair, and other functions	Many types of cancer
<i>RB</i>	Regulates cell division	Retinoblastoma and many other cancers

Tumor suppressor genes are often inherited as a dominant trait.

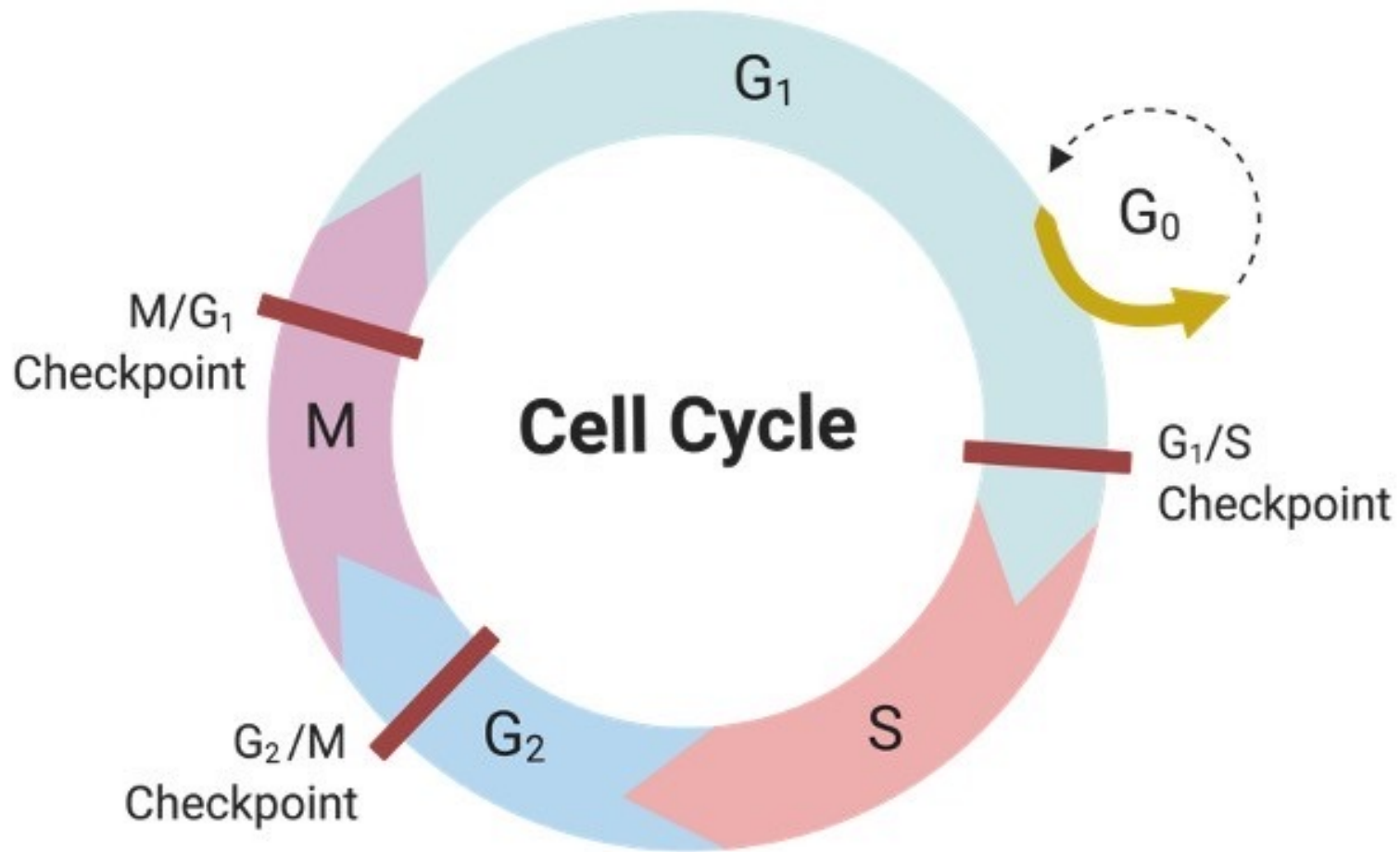
Pedigree structure of a family with familial adenomatous polyposis (APC).

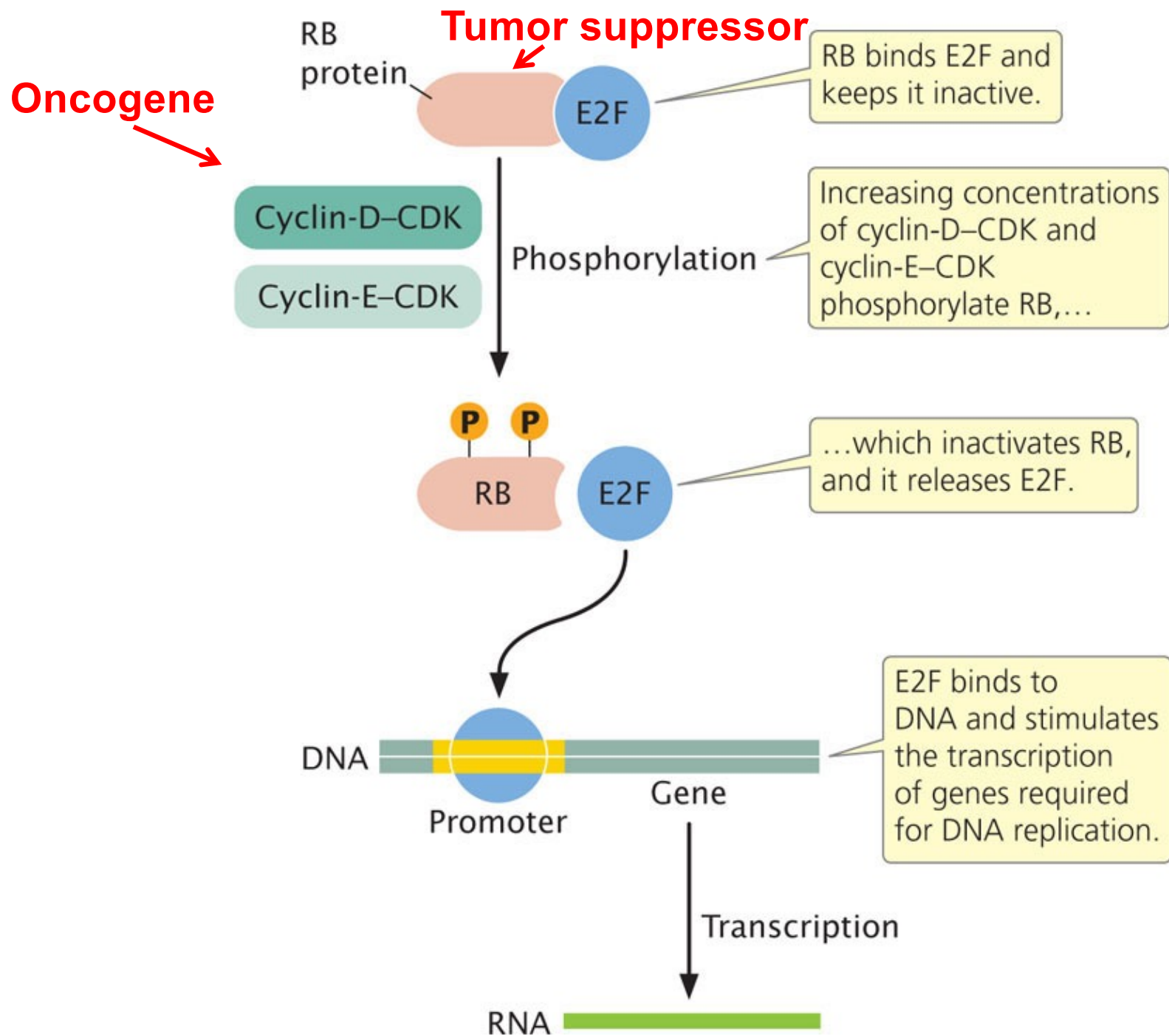


-The chance of acquiring the second mutation in a somatic cell is high. In theory, a single cell acquiring the second mutations is sufficient to cause cancer.

-Not all carriers will develop cancer (penetrance)

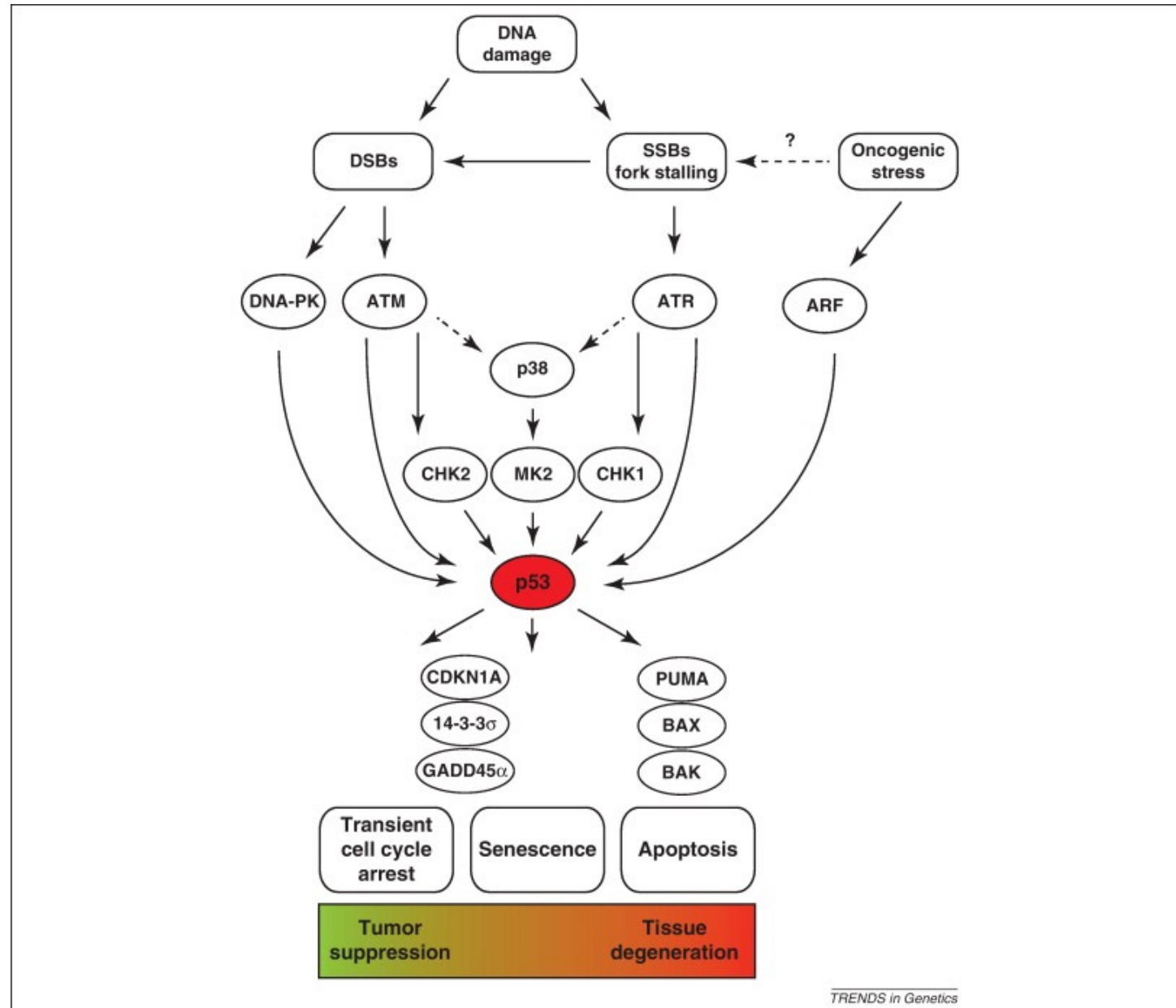
The cell cycle checkpoint





Pierce Figure 23.7

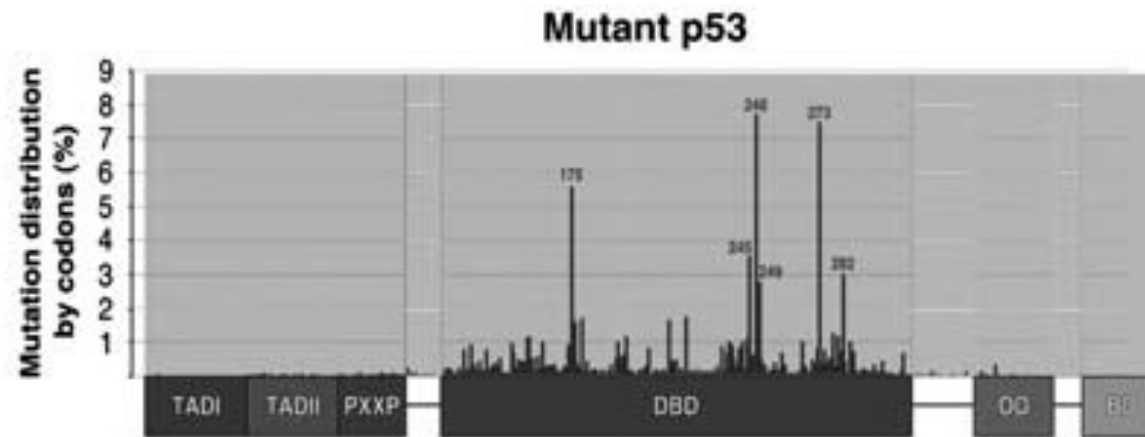
The guardian of the genome : p53



The guardian of the genome : p53

p53 mutation is inherited in Li-Fraumeni syndrome.
p53 is mutated in > 50% of all human tumors.

The p53 protein



Goldstein et al. 2011

TAD : transcription activation domain

DBD : DNA-binding domain

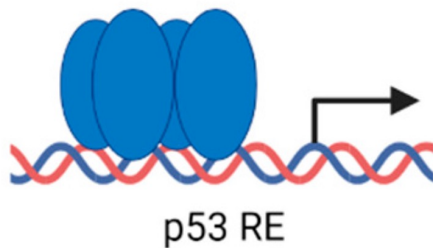
OD : oligomerization domain

Unable to bind DNA but able to form tetramer!

Many p53 mutant alleles in cancer function as a dominant negative

P53 is a transcription factor that binds DNA
as a homotetramer

wild-type

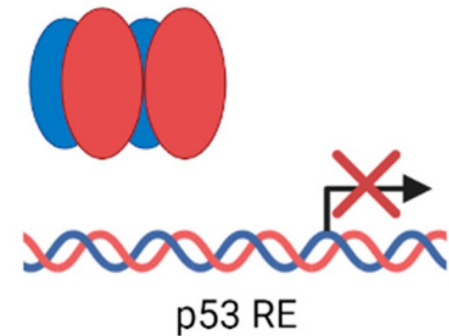


Activation of canonical
p53 target genes

Mutation in the DNA
binding domain in one
of the two alleles



Dominant-negative effect



No activation of canonical
p53 target genes

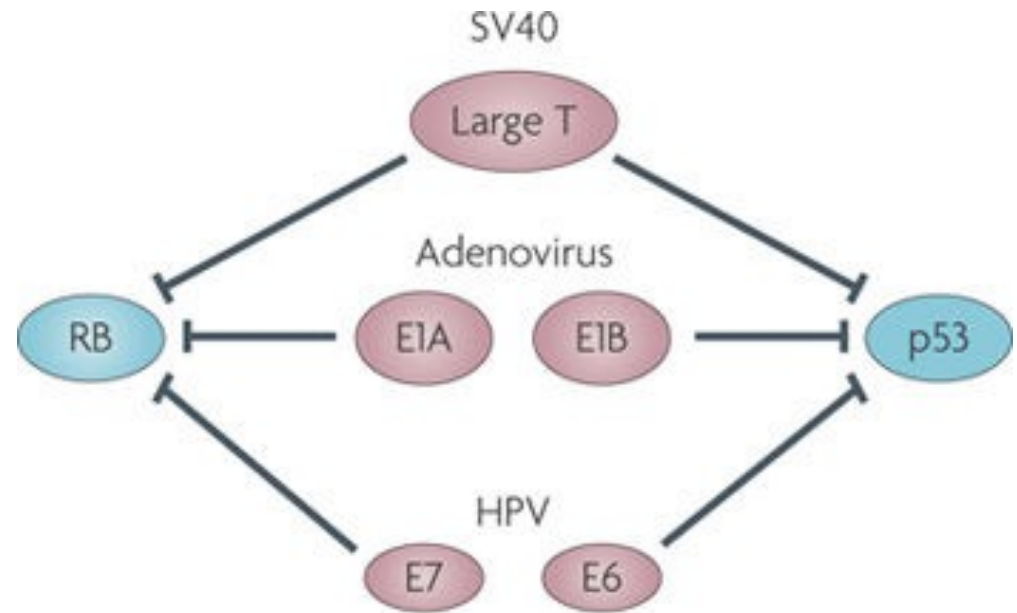
Table 23.6 Some human cancers associated with viruses

Virus	Cancer
Human papilloma viruses (HPVs)	Cervical, penile, and vulvar cancers
Hepatitis B virus	Liver cancer
Human T-cell leukemia virus 1 (HTLV-1)	Adult T-cell leukemia
Human T-cell leukemia virus 2 (HTLV-2)	Hairy-cell leukemia
Epstein–Barr virus	Burkitt lymphoma, nasopharyngeal cancer, Hodgkin lymphoma
Human herpes virus	Kaposi sarcoma
Merkel cell polyomavirus	Merkel cell carcinoma

95% of cervical cancer patients are infected with HPV

Viral oncoproteins from DNA tumor virus

Viral oncoproteins from DNA tumor virus bind and inactivate Rb and p53

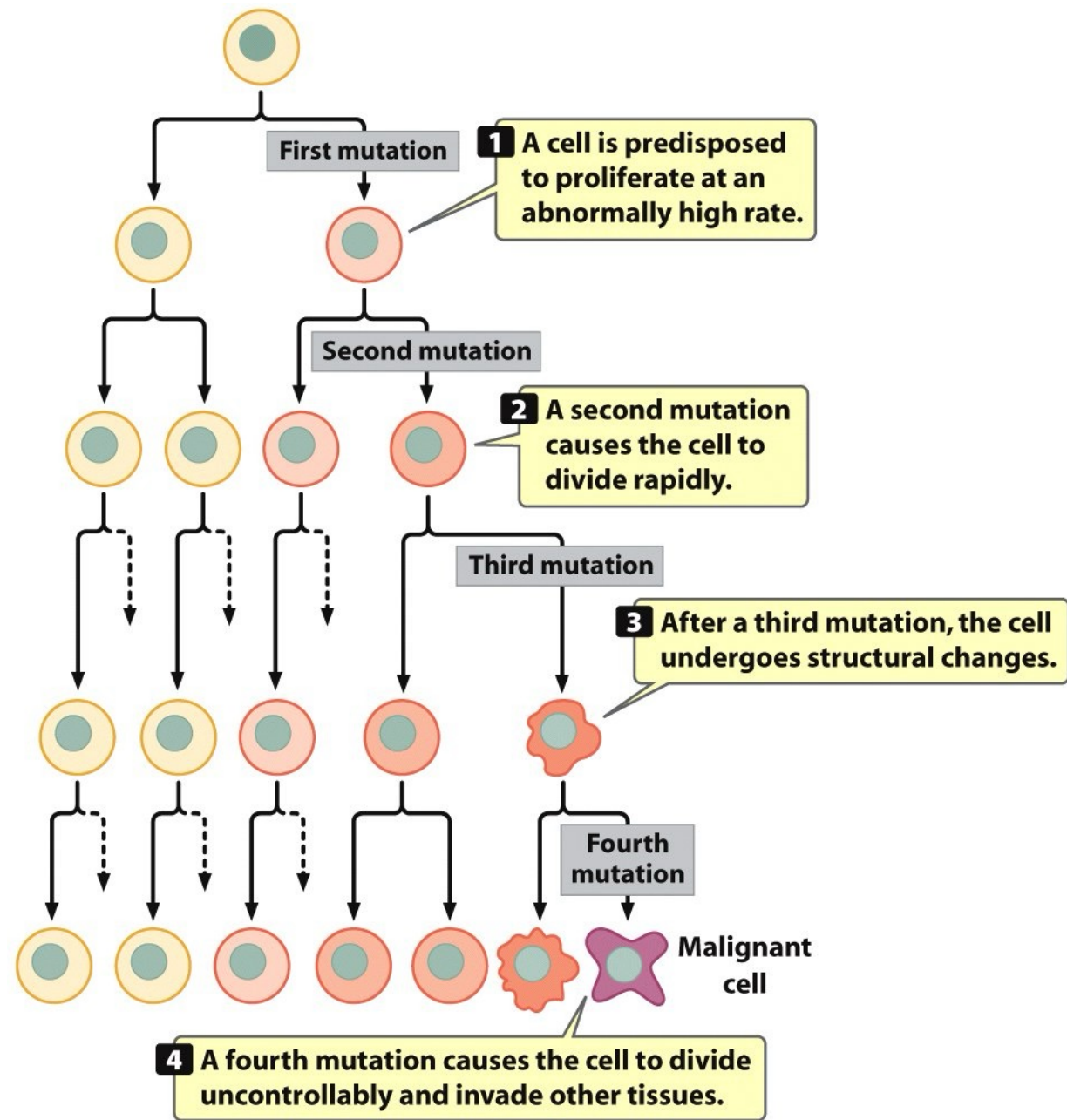


Clonal evolution of cancer cells

A single cell acquiring a mutation that provides proliferative advantage gives rise to clones of mutated cells.

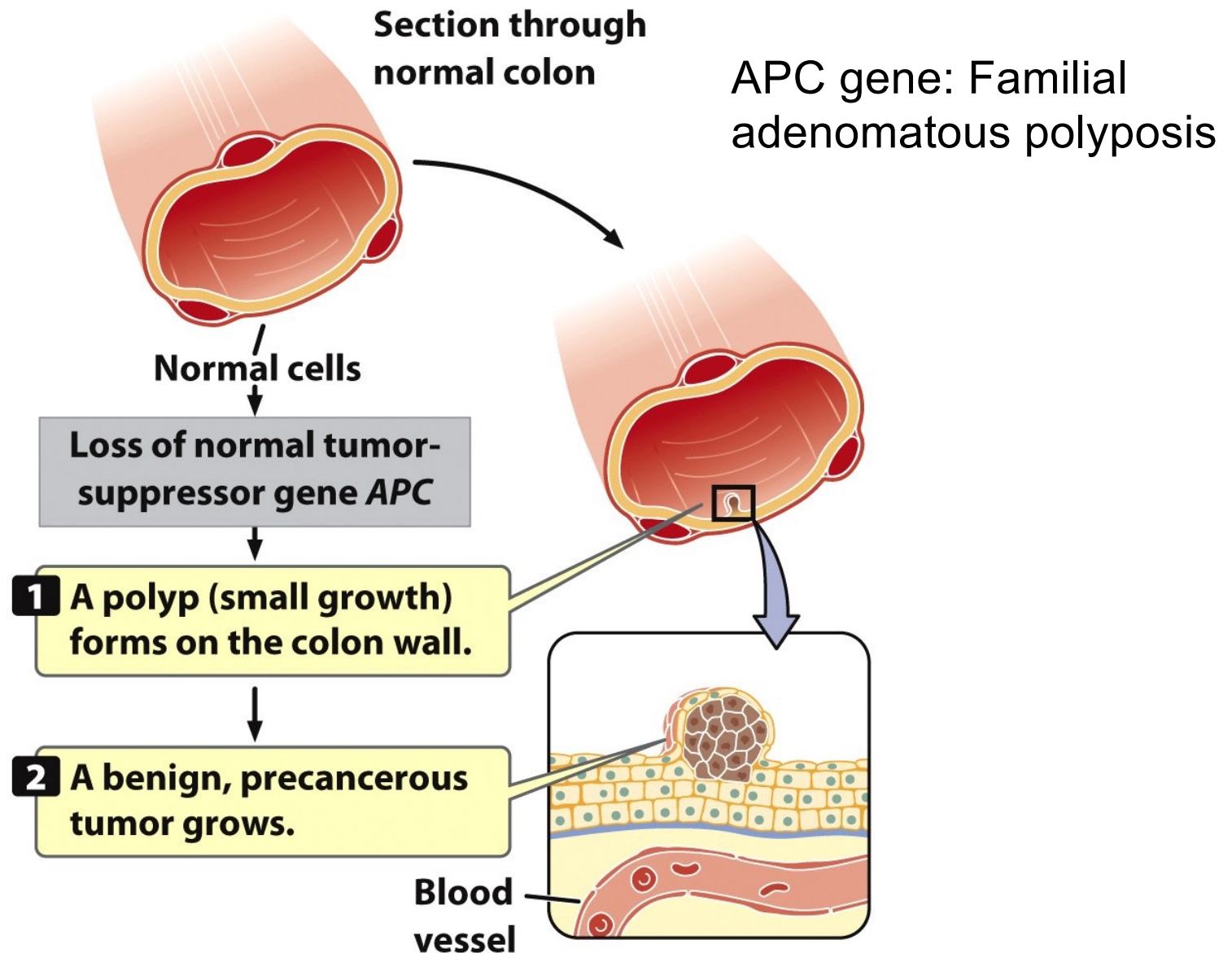
A single cell from the clones may acquire a secondary mutation that provides additional proliferative advantage.

Eventually, cycle of acquiring mutations and clonal expansion gives rise to fully transformed cancer cells



Pierce Figure 23.4

Defined genetic changes are associated with specific stages of colon cancer development



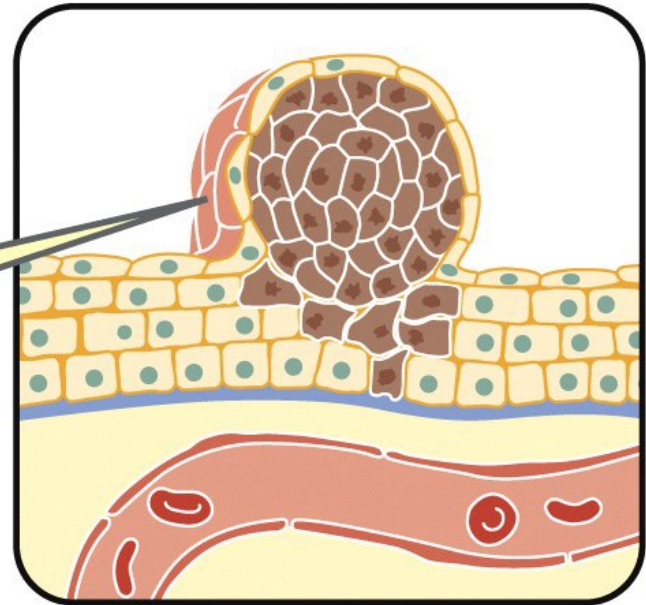
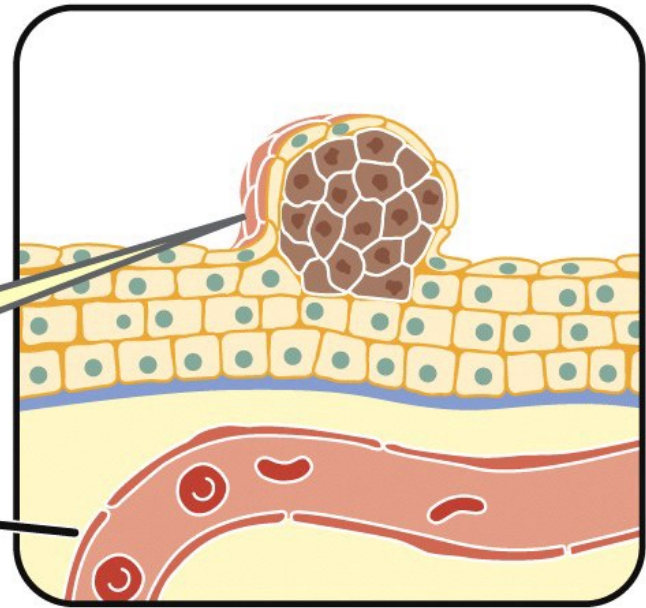
Pierce Figure 23.10

2 A benign, precancerous tumor grows.

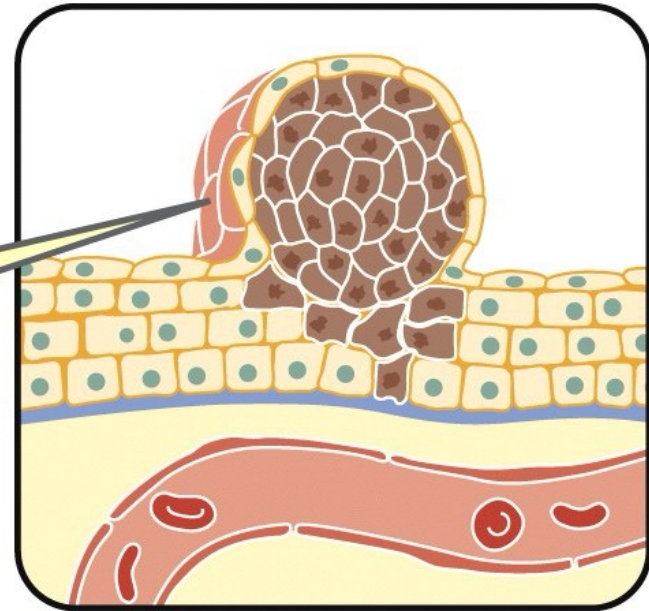
Activation of
oncogene *ras*

3 An adenoma
(benign tumor) grows.

Blood
vessel

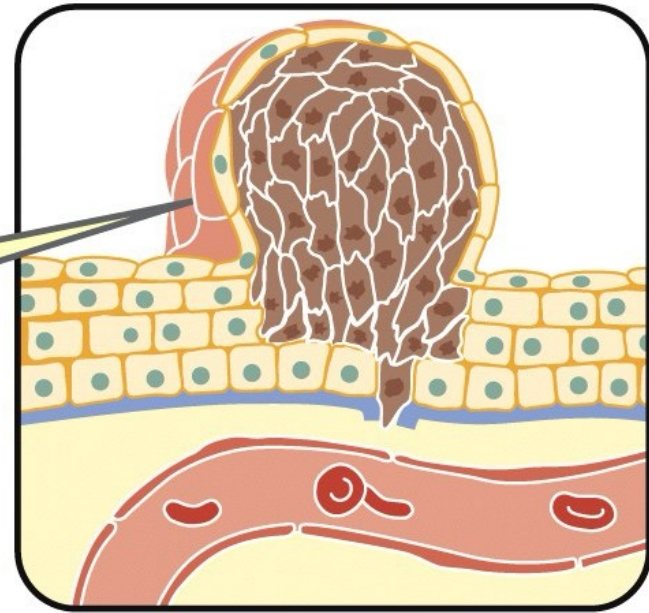


3 An adenoma (benign tumor) grows.

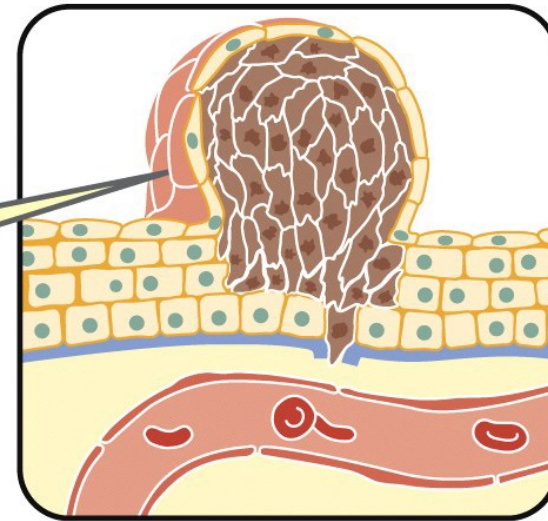


Loss of tumor-suppressor gene *p53*

4 A carcinoma (malignant tumor) develops.

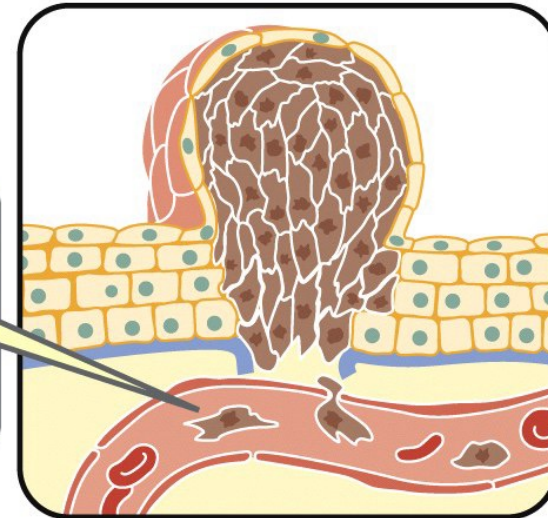


4 A carcinoma (malignant tumor) develops.



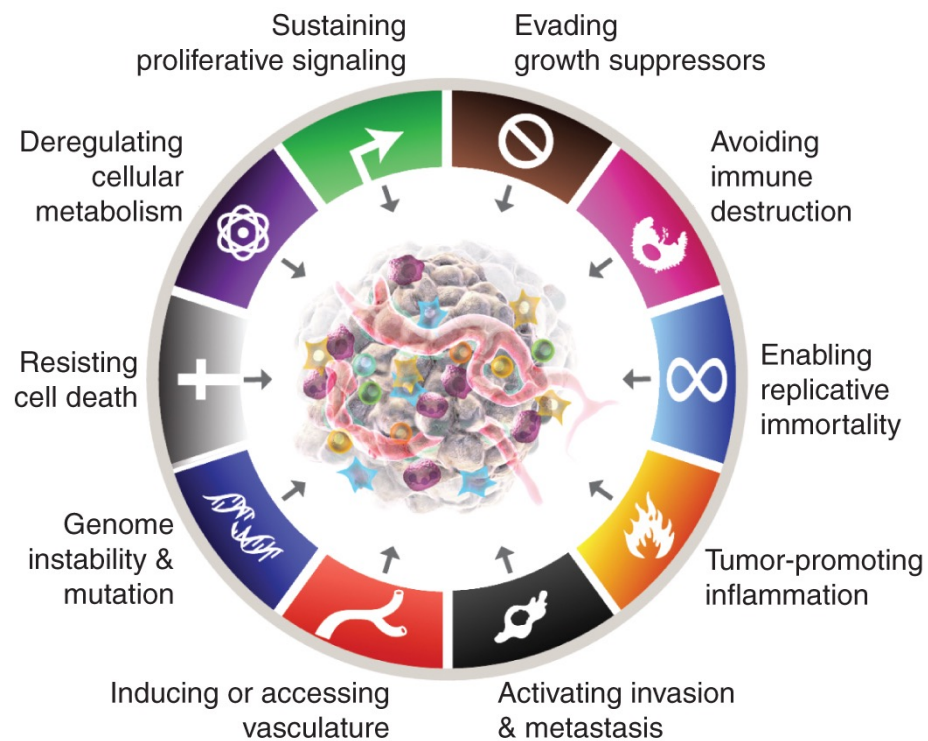
Other changes; loss of antimetastasis gene

5 The cancer metastasizes (spreads to other tissue through the bloodstream).

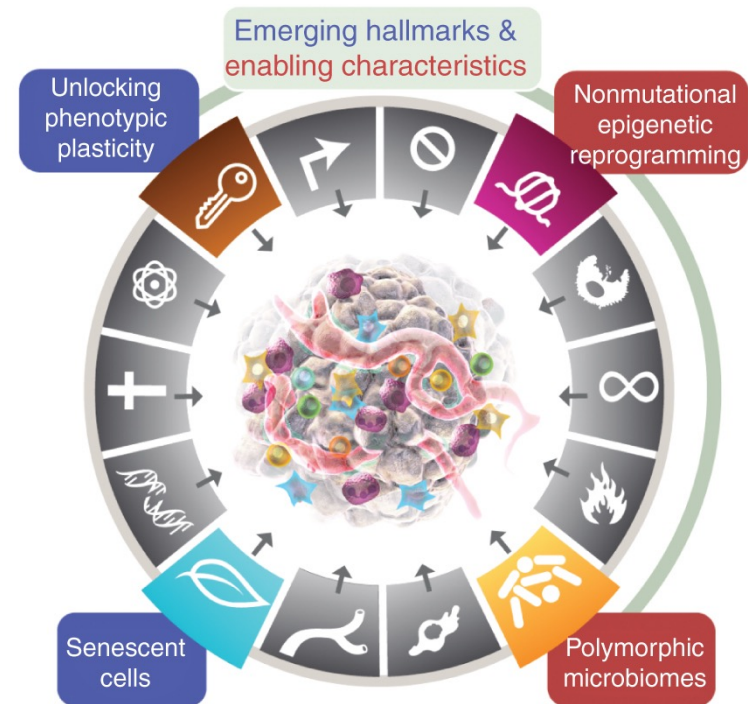


Cancer is a multi-step process, involving many genetic alterations (not necessarily in the same order).

There are still a lot to learn about cancer, Hallmarks of Cancer II & III

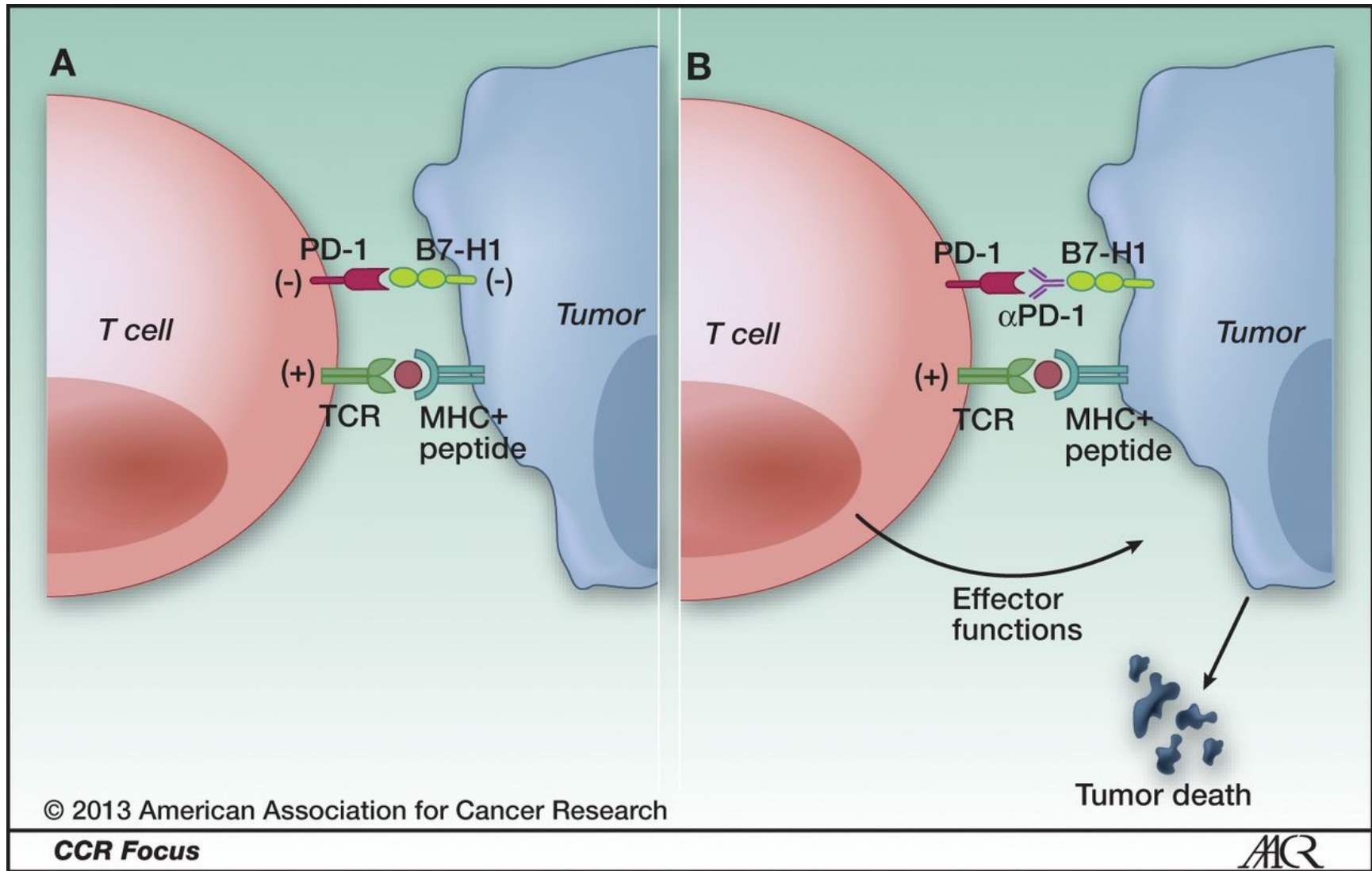


2011

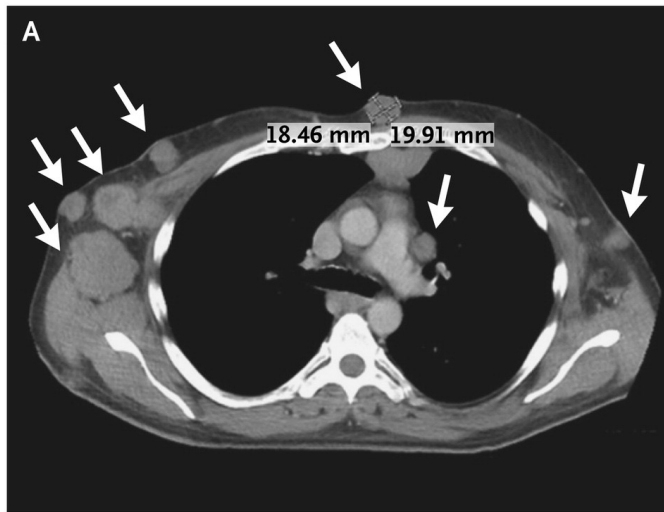


2022

PD-1 activation by tumor cells suppresses T cell function



Chest showing tumor regression in a patient treated with CTLA-4 and PD-1 antibodies



Before



4 weeks after treatment

Role of environmental factors in Cancer

Table 23.3 Percentage of cancer cases in the United Kingdom caused by environmental factors

Factor	Percentage of Cancer Cases
Tobacco	19.4
Diet	9.2
Overweight and obesity	5.5
Alcohol	4.0
Occupation	3.7
Radiation (UV)	3.5
Infections	3.1
Radiation (ionizing)	1.8
All environmental factors	42.7

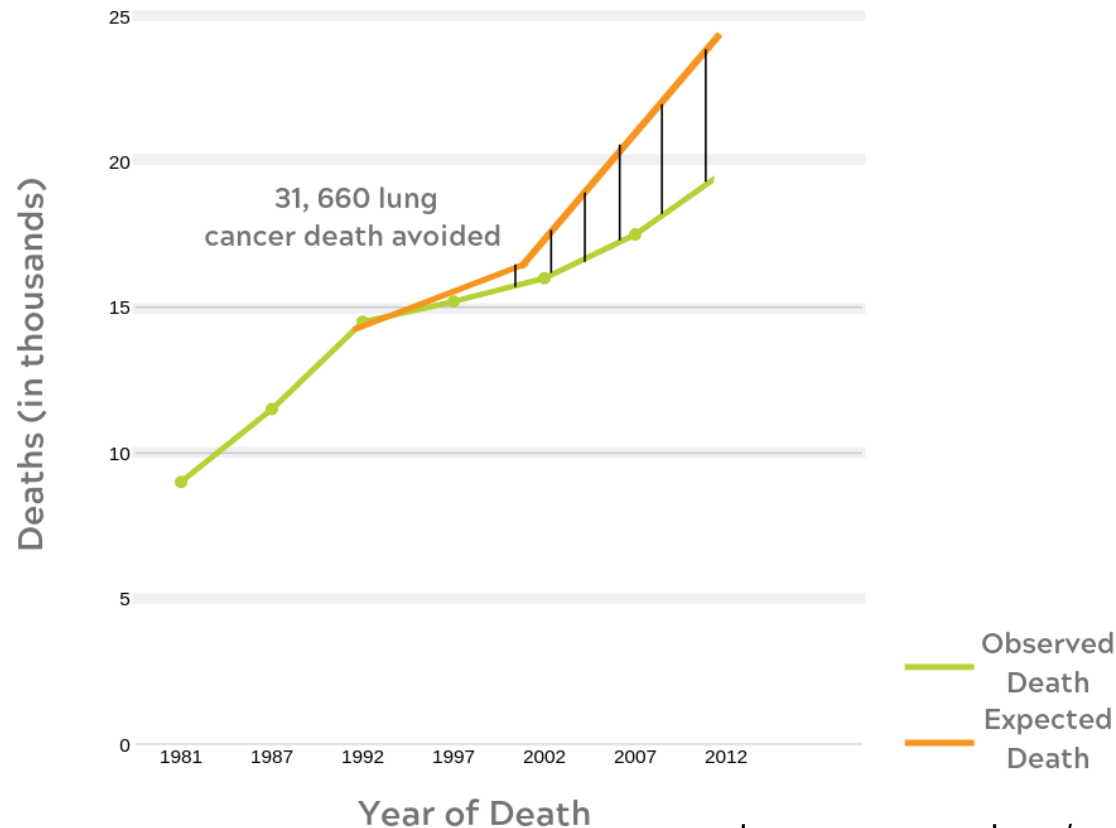
Source: Parkin, D.M., L. Boyd, and L.C. Walker. 2011. Fraction of cancer attributable to lifestyle and environmental factors in the UK in 2010. *British Journal of Cancer* 105:S77-S81.

Table 23.3

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Role of environmental factors in Cancer

Lung Cancer, 1981-2012



www.lungcancercanada.ca/

-The U.S. ban on in-flight smoking began with domestic flights of two hours or less in April 1988. All planes became smoke-free by the end of the 1990s.